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October 11, 1966

INTERIM FEDERAL SPECIFICATION

WRENCH, BUNG, AND WRENCH, DRAIN PLUG

This Interim Federal Specification was developed by Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C., 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wrenches for use on drum bungs, and various sizes of female drive drain plugs.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the title of the specification, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types. The wrenches shall be of the following types as specified (see 6.1):

Type I. - Bung.

Type II. - Drain plug.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:
Fed. Std. No. 123 - Marking for Domestic Shipment.
Fed. Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20406.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:
MIL-H-15424 - Hand Tools; Packaging of.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of wrenches which are otherwise in accordance with this specification.

3.2 Material. Material shall be as specified herein. Material shall be free from all defects and imperfections that might affect the serviceability and appearance of the finished product.

3.3 Type I wrench.

3.3.1 Design. Type I wrench shall be capable of loosening and tightening bungs on metal drums. It shall be of the single-end, or double-end, multiple-wrenching, engager type. The wrench shall be capable of engaging at least 20 different common commercial bungs used on storage drums, including round, square, hexagon, channel, and lug form, male and female bungs. The minimum length of the wrench shall be 15 inches. Wrenches shall be similar to figure 1.

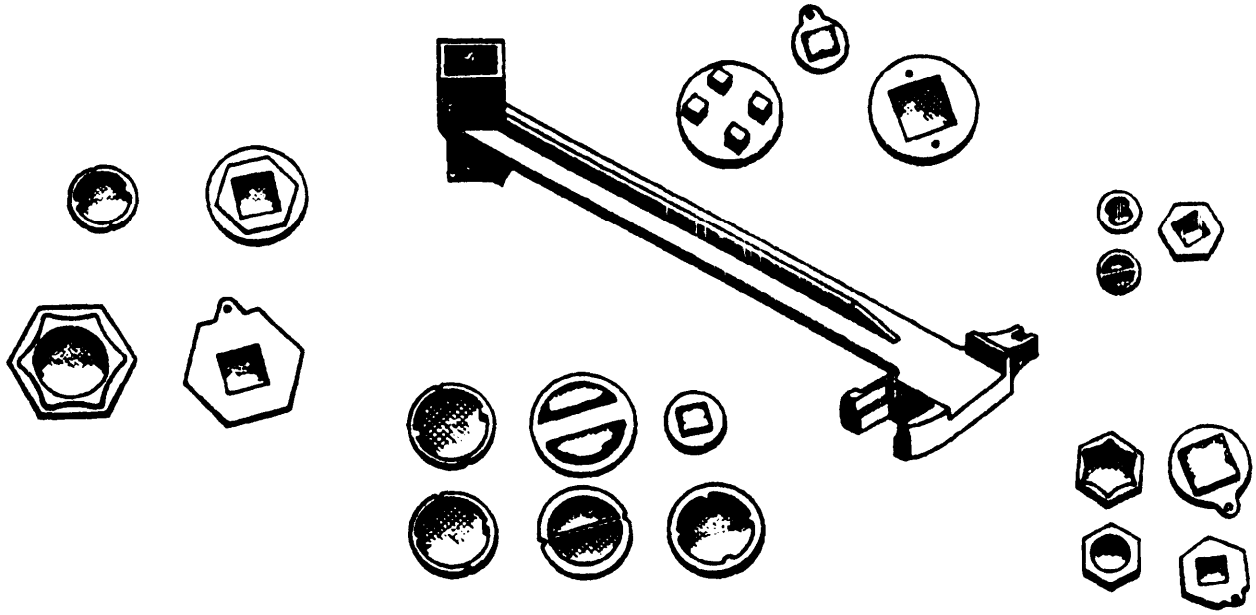


Figure 1. Type I, bung wrench (showing 20 common types of bungs which wrench will fit).

3.3.2 Heads. Type I wrench head(s) shall be constructed of steel or malleable iron and have a Rockwell hardness of not less than 35 nor more than 55 on the C scale when tested in accordance with 4.4.4.

3.3.3 Handles. Type I handles shall be of steel and securely attached or integrally forged to the wrench head(s) so as to provide a balanced tool. Wrench handles shall clear the rim of the barrel head with a minimum clearance of one inch when engaged in loosening or tightening bungs.

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3.3.4 Test loads. Type I wrenches shall withstand a minimum test load of not less than 6100 inch-pounds for the minimum bung size and a maximum test load of 11,000 inch-pound for maximum bung size without injury or permanent deformation (set) which may affect their durability or serviceability (see 4.4.2).

3.4 Type II wrench.

3.4.1 Design. Type II wrenches shall be of the double-end type with multiple-wrenching engagers for loosening and tightening various sizes of female drive drain plugs. Each wrench shall include among its various drain plug sizes, square engagers sized 5/16, 3/8, 7/16, 1/2, 9/16, and 5/8 inch. Tolerances for engager sizes shall be plus 0, minus 1/32 inch. In addition, the wrench shall have a screwdriver-type blade with a thickness of 1/8 inch at the tip ± 1/64 inch and 3/4 inch in width at the tip, ± 1/8 inch. The wrench shall be 8-3/4 inches overall length ± 1/4 inch and shall be similar to figure 2.

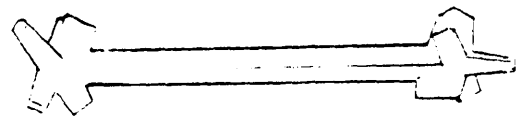


Figure 2. Type II, drain plug wrench.

3.4.2 Material. Type II wrench shall be of forged steel and have a Rockwell hardness of not less than 40 nor more than 45 on the C scale when tested in accordance with 4.4.4.

3.5 Finish. Wrenches shall be painted or otherwise treated to resist corrosion in accordance with the manufacturer's standard practice. They shall be free from scale, pits, burrs, cracks, sharp edges or other detrimental defects that might affect serviceability.

3.6 Marking for identification. Wrenches shall be marked in a legible and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table I.

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Table I. Sampling data.

Category	Sample Unit	Inspection Level	Acceptable Quality Level	AQL Expressed in Terms of	Reference
Visual examination	1 Ea.	II	4.0	Defects per hundred units	4.3.1
Dimensional examination	1 Ea.	S-4	1.5	do.	4.3.2
Testing	1 Ea.	S-4	1.0	do.	4.4
Preparation for delivery	One container	S-2	4.0	do.	4.5

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1, 4.4.2, 4.4.3, and 4.4.4.

4.4.1 Preparation for load test. Straight lines, approximately parallel, shall be scribed the full length of principal axes of the wrench. It is suggested that the sample wrench be securely fixed in a definite position on a vertical faceplate, and the straight lines scribed in it by means of a surface gage. The external dimensions of the wrench engagers shall be measured. Suitable grips and sockets shall be supplied to fit into wrench openings and on engagers to provide reaction against the load. All mandrels and sockets shall be hardened to a Rockwell hardness of between 40 and 46 on the C scale.

4.4.2 Application of load. Wherever possible, the load shall be applied at the center of the natural handgrip on the handle, or as far away as practicable from the opening, engager, or slot of the wrench. A shallow V-groove shall be filed on the wrench handle to mark the point of load application. The load shall be applied through a knife-edge bearing fitted into the V-groove and shall be held for one minute. The load shall be applied with suitable torque-producing machines, by means of dead weights or testing machines. The effective lever arm, when the load is vertical and lever is employed, is the shortest distance in inches between two vertical lines passing respectively through point of load application and the center of the reaction mandrel or socket. The specified test load shall be the product of the effective lever arm, in inches, and the applied load, in pounds (see 3.3.4).

4.4.3 Determination of test results. After application of the load test, the test wrenches shall be replaced in the same position on the vertical base plate. The scriber point on the surface gage shall be moved along the scribed lines on the wrenches. Any permanent deformation of handles will be indicated by a change in the straight reference line scribed on the wrench before application of the load. Sockets, slots, and engagers shall be remeasured and examined. Any evidence of permanent set in the handles; distortion or spreading of the slots, engagers, or openings; or other indication of weakness of any nature that would affect the serviceability of the wrench shall be cause for rejection.

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4.4.4 Hardness test. A hardness test shall be conducted in accordance with Fed. Std. No. 151 to determine compliance with 3.3.2 and 3.4.2.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Table II Classification of preparation for delivery defects.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any componert missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.1), the wrenches shall be preserved, packaged, and packed in accordance with MIL-H-15424. The level of preservation and packaging shall be A or C, and the level of packing A, B, or C as specified (see 6.1).

5.1.1 As a part of the requirements of section 5 of this specification, table III is considered a part of table I of MIL-H-15424.

Table III.

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
Wrench:					
Ring	P-1	I	1	—	12
Drain Plug	P-1	I	1	—	24

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Type (see 1.2).
(c) Selection of level of preservation, packaging, and packing required (see 5.1).

6.2 Civil agency packaging. When level A packaging is required for civil agency procurements, the level A requirements in 5.1 shall apply.

6.3 For nonsparking and nonmagnetic bung wrenches, see Federal Specification OOG-W-6h4.

6.h This specification includes, in part, the requirements of Federal Specification OOG-W-6h4, dated June 23, 1961.

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